Practice — Pure Substances and Elements

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 · MCQ **[1]**

An element is a pure substance that —

(a) can be split physically (b) cannot be broken into simpler substances (c) is always a mixture
(d) has two kinds of particles

Q2 · MCQ **[1]**

The smallest particle of an element is an —

(a) atom (b) alloy (c) mixture (d) compound

Q3 · MCQ **[1]**

Which set are all non-metals?

(a) iron, gold, silver (b) carbon, sulfur, oxygen (c) silicon, boron (d) copper, zinc, tin

Q4 · MCQ **[1]**

Silicon and boron, with in-between properties, are called —

(a) metals (b) non-metals (c) metalloids (d) alloys

Q5 · ASSERTION–REASON **[1]**

A: Water is not an element.
R: Water can be broken into hydrogen and oxygen.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK **[1]**

Two or more atoms joined together form a ________.

____

Q7 · FILL IN THE BLANK **[1]**

A pure substance is made of only one type of ________.

____

Short answer (2 marks each)

Q8 **[2]**

What is a pure substance in science? How does it differ from the everyday meaning of 'pure'?

Q9 **[2]**

Define an element and give two examples each of metals and non-metals.

Q10 **[2]**

Differentiate between an atom and a molecule.

Long / application (3 marks each)

Q11 **[3]**

Classify each as element, and as metal/non-metal/metalloid: gold, oxygen, silicon, sulfur. Explain what 'element' means.

Q12 **[3]**

A food packet says 'pure ghee'. Explain how a scientist's idea of 'pure' could differ, and why the label can still be acceptable.

Check yourself → Answer key